

Behavioral and Brain Sciences

A speculative argument against consciousness in AI (and perhaps some invertebrates)

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Abstract:	Brain waves driven by electrochemical processing may be necessary for consciousness, and if so, that will challenge consciousness in AI (and perhaps in some invertebrates).

A speculative argument against consciousness in AI (and perhaps some invertebrates)

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ABSTRACT

Brain waves driven by electrochemical processing may be necessary for consciousness, and if so, that will challenge consciousness in AI (and perhaps in

some invertebrates).

MAIN TEXT

See below.

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I will sketch one line of speculative support for Seth's thesis of substrate-dependence of consciousness.

It has long been recognized that brain rhythms are important in neural functioning (Kopell, Gritton, Whittington, & Kramer, 2014). At the dawn of the connectome era, Kopell, et al. suggested that we think not in terms of a static connectome but rather dynamically, in terms of the “dynome”, and the key to the dynome is brain rhythms. Recent evidence suggests that electrochemical mechanisms may be required for brain rhythms and that brain rhythms are required for sophisticated processing. If—and this is highly speculative -- they are required for consciousness itself, that would preclude consciousness in non-biological AI systems.

The connectome project has strikingly confirmed the widely held view that individual neurons, especially in the vertebrate cortex tend to have so many synapses on each of their axons that pairwise synapses are too weak to explain post-synaptic neural firing (Fields, 2025; Shapson-Coe et al.). R. Douglas Fields puts this point as follows (2025, p. 5): “Ironically, the goal of the brain initiative to trace out the connectome of the human brain has revealed that there is none”. More specifically, in the vertebrate cortex, most synapses are “en passant”, axons have many synaptic offshoots along the length of the

axon. The result is that the activity in these synapses “are too weak to depolarize the postsynaptic neuron to the action potential threshold” (Fields, 2025, p. 1).

Instead, neural firing in the cortex depends on the joint action of populations of neurons themselves partly driven by causal interactions outside the neurons. For example, there are waves of ion fluctuations in the chemical soup surrounding neurons. Every charged particle generated by a neuron has to return via this soup. And glial cells regulate ion concentration in the soup and absorb and release neurotransmitters. These factors together with subthreshold oscillations in cell somas, dendrites and axons together with ionic oscillations in the soup create powerful population-wide electrical waves that are part of what drives the firing of neurons (Fields, 2025). These waves make up for the weakness of direct pairwise connections between neurons.

As Peter Godfrey-Smith notes (2024), if we favor gradualism in evolution, one place to look for clues to the nature of consciousness would be the evolutionary origins of consciousness. The first animals appear to have been comb jellies (Schultz et al., 2023). What Schultz, et al. showed was that comb jelly chromosomes were radically different from those of subsequent animals, including sponges and us. Comb jellies have neurons but their immediate successor, sponges, do not. Comb jellies might have been on a dead-end evolutionary branch. Or else sponges evolved from comb jellies but lost their neurons. Either way, comb jelly neurons seems didn’t lead to more advanced systems. Burkhardt, et al. (2023) showed that the comb jelly nervous system (of at least one phase in their lifecycle) was entirely electrical. As *Scientific American* (Giovanetti, 2024) asked, “Did Neurons Evolve Twice?” The answer is yes in both scenarios, so it looks like electrochemical processing (in which neurotransmitters are released by one neuron and trigger electrical activation in another neuron) has a functional advantage over purely electrical processing.

As Seth notes, dynamical approaches to consciousness leave open the question of whether the substrate matters and why. In this case, the issue is whether there is something about the electrochemical implementation of these brain rhythms that is crucial to consciousness. The new results don’t answer that question but they do suggest that electrochemical operation may have been superior in one or another way that aeons later led to sophisticated vertebrate brains.

Even if electrochemical processing is necessary for consciousness in actual creatures, and is “mortal” in the sense of requiring a certain substrate in actual creatures, that leaves open *why* that is so and does not entail what Seth calls “strong” substrate dependence in which consciousness is supposed to depend on “intrinsic” properties of its material basis.

The argument so far is: sophisticated cortical processing requires brain rhythms. Brain rhythms depend on electrochemical processing. Speculation: consciousness in us depends on electrochemical processing. Further speculation: consciousness tout court depends on those mechanisms.

Here is another layer of speculation: Brain rhythms may depend on low frequency oscillations (Bullock & Basar, 1988). Notably, those low frequency oscillations are more prominent in vertebrates than in many invertebrates. One kind of glial cell, the oligodendrocyte, modulates myelin affecting low frequency neuronal firing (Pajevic, Plenz, Bassar, & Fields, 2023). Both oligodendrocytes and myelin are unique to vertebrates. The low frequency waves spread much more widely than high frequency waves, on the order of centimeters for under 20 Hz but at the micrometer level for over 50 Hz. Although there are low frequency brain waves in invertebrates (Marder, Goeritz, & Otopalik, 2015), many invertebrate neural systems (with the notable exception of the octopus (Bullock & Basar, 1988)) are said to be weak in highly coherent low frequency brain waves (Fields, 2025) and in some, especially more primitive invertebrates, consist mainly in strong serial connections with much less need for the brain waves. See Figure 1.

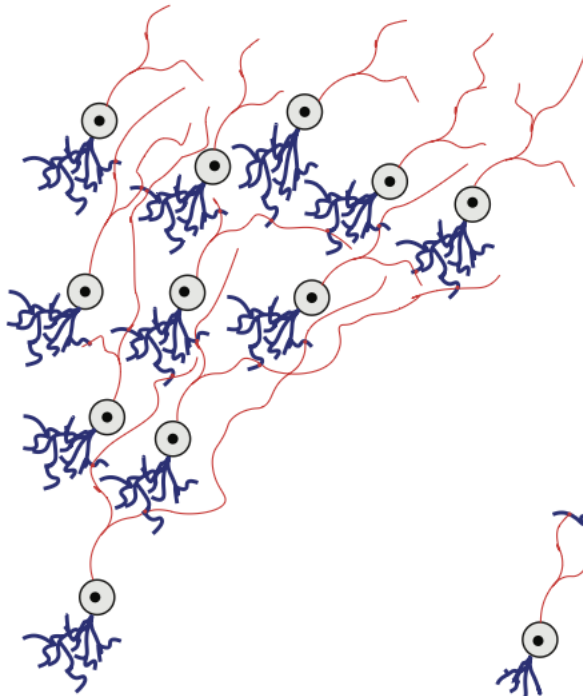
However, there are slow waves in NREM sleep and under anesthesia. Siclari, et al. (2018) distinguish two types of slow waves. NREM dreams take place in a background of slow waves, but are associated with decreases in the delta range in one type of slow wave in central and posterior cortex together with increases in the other type of slow wave in frontal areas. Slow waves are present in anesthesia (Pazienti, Galluzzi, Dasilva, Sanchez-Vives, & Mattia, 2022) but I know of no data on which of the two kinds of slow waves is present.

In sum, there is an intriguing but empirically motivated speculation that argues against consciousness in AI and a further speculative line of thought that may exclude consciousness in some invertebrates.

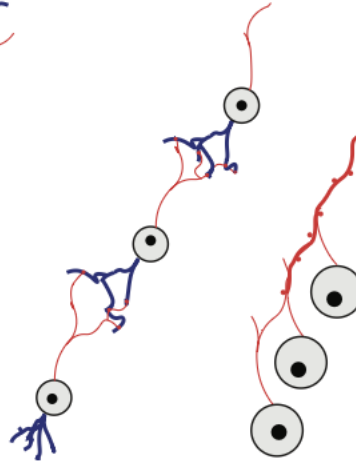
Figure caption

Figure 1. In vertebrates, axons (red lines) wander through a forest of dendrites (blue lines), with many synapses. The circles with dots are the cell bodies. There is a tendency in invertebrates towards synapses that are stronger because axons link to only a few dendrites or even one dendrite. From (Fields, 2025), reprinted by permission of Sage Publications.

Human cerebral cortex



Invertebrates

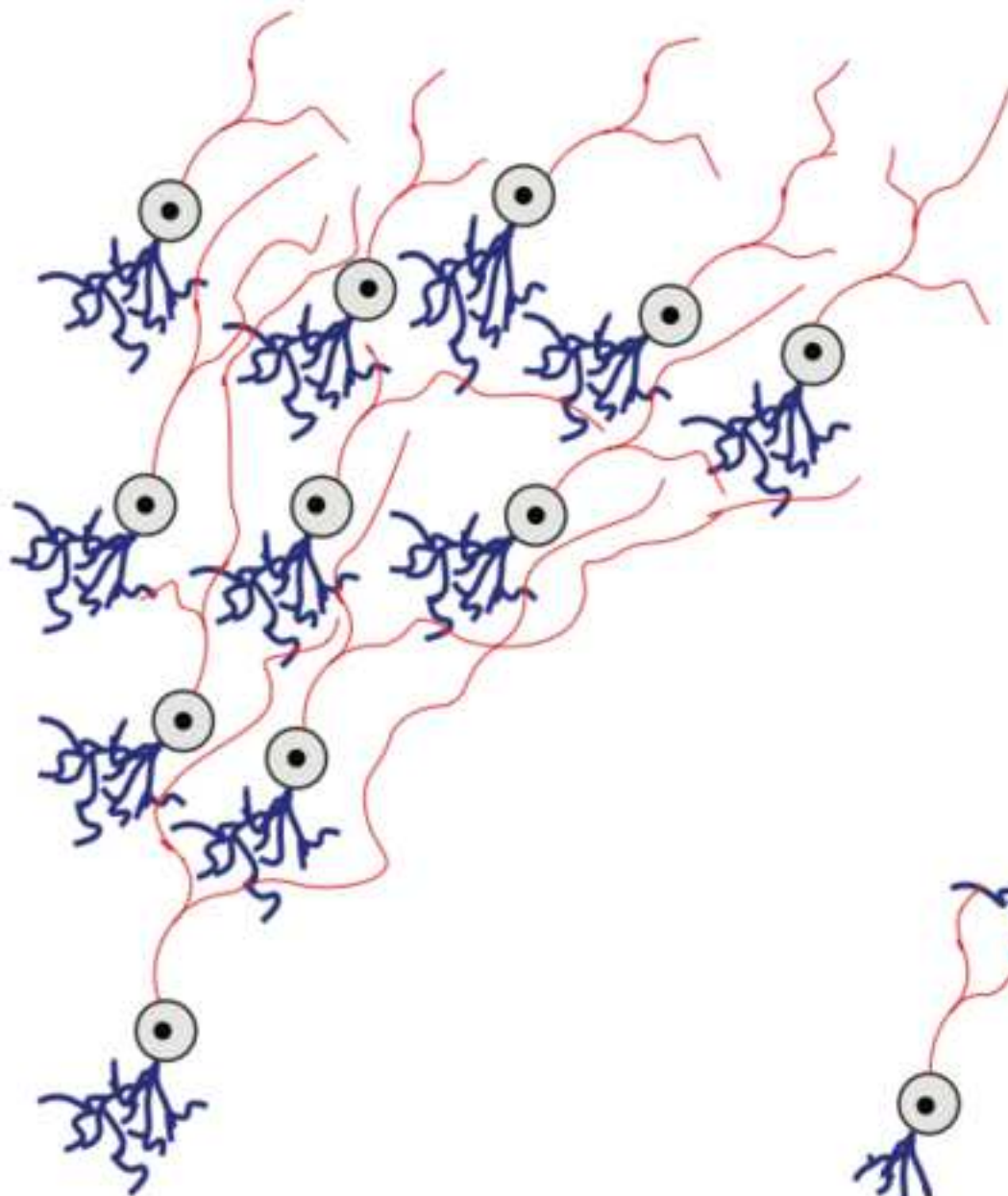


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Human cerebral cortex



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